

VisTemp - Developing Visual Temperature Prediction from Semantic Segmentation of Street View Imagery

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Abstract

This study is focused on predicting air temperature on street level using the CityScape dataset (Cordts et al., 2016), which contains 5,000 finely annotated images for 8 categories, complete with metadata related to date, location and outside air temperature, collected from 50 European cities. We present a neural network architecture that is trained on the CityScape dataset for dual purposes: segmenting urban images and predicting air temperature. Our network is based on a modified U-Net architecture with a ResNet encoder and an Atrous Spatial Pyramid Pooling (ASPP) after the bridge. The network has two heads: one for segmentation, capturing semantic data of urban scenes, and another for predicting the outside air temperature, informed by subtle features extracted from urban imagery. This structure enables a streamlined exploration of the relationship between various urban features and their impact on ambient temperature, contributing to our understanding of urban heat islands. Our Model performs reasonably well, given the vastly diverse input data we reached a mean IoU well above 0.7. The temperature prediction reaches an RMSE of 5.6 °C. The key finding is that the balancing of hyperparameters for optimizing the output accuracy for both tasks remains a challenge.

Github:

<https://github.com/matzeft/Streetview-Image-Decoding.git>

1. Introduction

The application of computer vision techniques to urban imagery offers numerous opportunities to investigate various socio-economic and physiological influences on the urban built environment.

Methods such as image segmentation, object detection, and scene classification algorithms enable in-depth analysis of urban areas. Studies combining such techniques with urban images have been successfully conducted to gain various insights, such as determining the proportion of urban greening [1], creating building typologies by

classifying buildings [2], and evaluating street spaces for bikeability [3].

A predominant method in such studies is to identify various indicators from street images through semantic segmentation. From this, indicators for the effective evaluation of urban landscapes are calculated. One notable example is the Green View Index (GVI), as developed by Li et al. [1] in 2015, which is defined through the proportion of pixels identified as greenery in the image. Chen and Biljecki [4] take a comprehensive approach by amalgamating multiple such indicators derived from the semantic segmentation of Street View images, to derive a scoring system for the quality of public open spaces. Among other objective indicators, they incorporate indices based on the proportions of various semantic classes within the overall image, including greenery (measured using the Green View Index as explained earlier), buildings, water bodies, sidewalks, and more.

Fewer CV studies deal explicitly with a prediction of urban air temperatures: Chu et. Al [5] proposed a CNN (Convolutional Neural Network) model to estimate air temperature based on single images, and a RNN (Recurrent Neural Network) to estimate the air temperature for the last image in a sequence. Both networks are trained on a dataset that contains images from 9 webcams in distinct locations throughout the US for a combined total of ca 53000 images. They reached an RMSE of 4.28 °C for their CNN model, and an RMSE of 2.82 °C for their RNN model.

Another example of translating images of urban scenery using CNN is given through Volokitin et. al. [6]. The authors use a VGG-16 Architecture trained on the same dataset as in [5], and achieve a RMSE 4.96 °C in their best model.

In this study, we aim to combine previous works by implementing a multimodal network with dual functionalities: direct encoding of RGB Street view images into ambient Air Temperature and segmenting images into various classes representing driving factors of urban air temperature. Earlier studies required two separate models to first segment the image and subsequently produce an air temperature estimation. The training for such models was conducted on data sets recorded under controlled conditions. [5] and [6] trained their models on a dataset with fixed camera positions at uniform recording intervals.

Our model combines both segmentation and temperature prediction and was trained on a comparatively smaller dataset containing images with more dynamic scenery.

2. Methods

2.1. Dataset

We utilize the Cityscape Dataset [7] to train, validate and test our model. This dataset is centered on the semantic understanding of urban street scenes, featuring detailed annotations for urban imagery from 50 European cities, recorded over various months and during good weather conditions. It provides fine annotations for around 5,000 images, focusing on 30 classes grouped into 8 categories, including humans, vehicles, constructions, nature and more. Each image comes with metadata containing the ambient air temperature at the time of capture. This dataset is particularly valuable for segmentation tasks due to its diversity, including a wide range of dynamic objects like pedestrians and cyclists, often in complex, cluttered scenes. Each image comes in a resolution of 720p. The training dataset contains 2975 images, the validation dataset contains 500 images, and the test dataset contains 1525 images. Fig 1 showcases the distribution of pixels in each class for each category, demonstrating the diversity of the dataset.

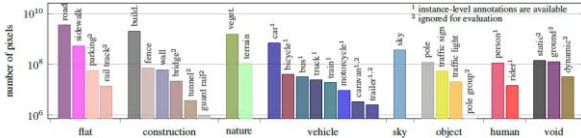


Figure 1: Original Cityscape classes distribution.

2.2. Data Preparation

The original Cityscape classes were further remapped into categories that are likely to influence the air temperature reading from the vehicle. The final 7 classes include Roads, Sidewalk, Built Environment, Vegetation, Sky, Vehicles, and Other. Built environment class is composed of classes building, wall, bridge, and tunnel. Vegetation class consists of both vegetation and terrain. Other features that are less likely to influence air temperature such as human, bicycle, traffic light, fence, and pole are grouped into others. Furthermore, not only does it relate the features to the air temperature, but the class remapping also reduces the unbalanced class size in the training data.

Other than the mask remapping, data augmentation for this model only includes resizing of the input images to 128

x 64 and RGB normalization. Outside temperature readings are kept as is.

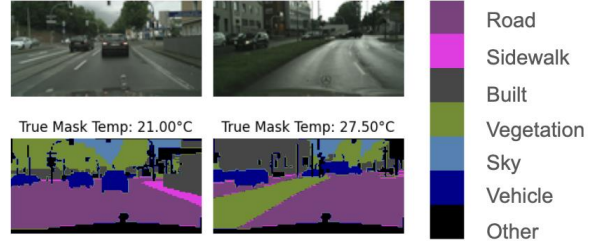


Figure 2: Remapped classes based on category that may influence air temperature.

2.3. Model Architecture

The proposed multimodal network is solemnly based on U-Net architecture with an added branch from the latent space. The encoder is based on ResNet 50 architecture [8], which applies multiple bottleneck layers consisting of a convolution block with a stride of 2 and identity blocks. The decoder is built on a single residual block with an input of transposed convolution with stride of 2, concatenated with the low-level features from the encoder. A key component of this network is the DeepLab Atrous Spatial Pyramid Pooling block [9] with a rate of 6, 12, 18. The use of different dilation rates allow the model to learn from different fields of view thus efficiently capture multi-scale contextual information. The temperature regression head is a fully convolution layer that includes a residual block to further reduce the dimension of the latent representation, to which then goes through an adaptive average pooling before flattening and applying linear units. The use of batch normalization and ReLu activation in the temperature regression head are applied to further improve the generalization between training and validation data.

Another important component in the model is the use of dropouts [10]. Initial training attempts produced models that heavily overfit to the training data that caused instability in the validation. Thus, dropouts are added at the end of each block in the encoder as well as in the temperature regression head with values of 0.3 and 0.7.

2.4. Loss Function

In the proposed model, the loss function combines cross-entropy and Mean Squared Error (MSE) to effectively handle both semantic segmentation and temperature prediction tasks. For the temperature prediction, MSE is adjusted with an alpha of 0.001. This approach prevents the unscaled temperature values from dominating the

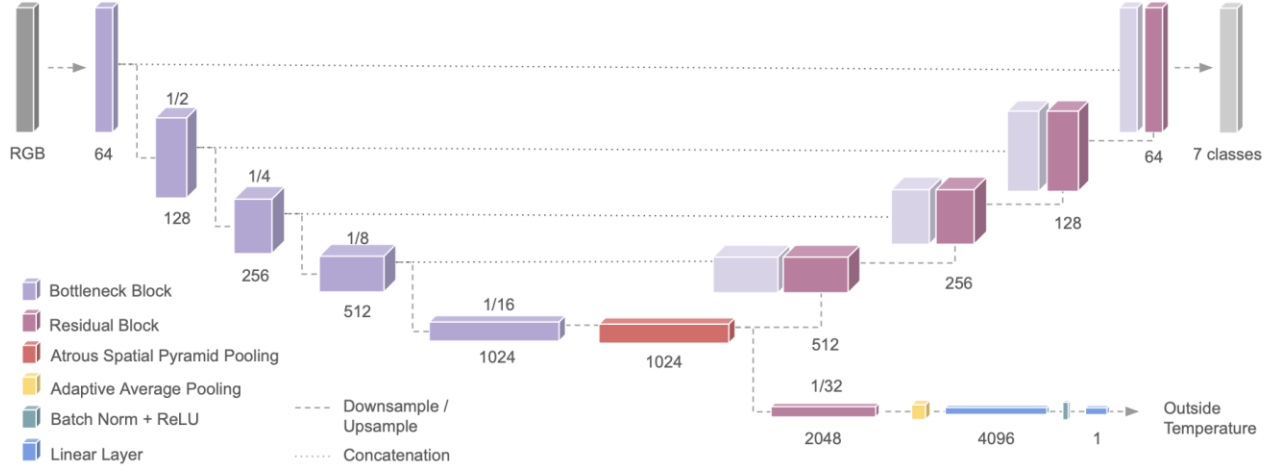


Figure 3: Proposed U-Net model with Atrous Spatial Pyramid Pooling and temperature regression head.

backpropagation process, thereby balancing the influence of both components in the training phase.

2.5. Training Setup

The proposed model was trained on 35 epochs with an AdamW optimizer and a cosine annealing learning rate scheduler that starts from $5e-4$ to $5e-5$. These values were determined empirically to balance both Cross-Entropy and MSE efficiently. Weight decay of $1e-3$ is applied to further reduce overfitting. Batch size is set to 32 with num_worker set to 16 to accelerate the training time. The model was trained on RTX 4090 GPU and took about 60 minutes to complete 35 epochs.

3. Results and Discussions

3.1. Initial Training

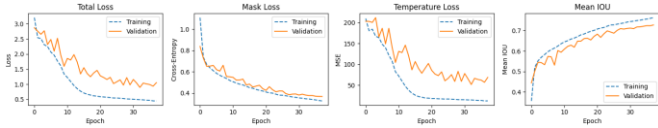


Figure 4: Loss Curves for initial test run with larger alpha and no gradient clipping.

Our initial training attempt was done with MSE scaling of 0.01 and no gradient clipping. These hyperparameters produced an unstable validation loss that is still dominated by temperature loss. Both mean IoU reached 0.7 where the RMSE is roughly 7.7 Celsius. A significant plateau was

observed in the loss after 15th epoch, suggesting that the minima has been missed and learning has become limited.

3.2. Final Results

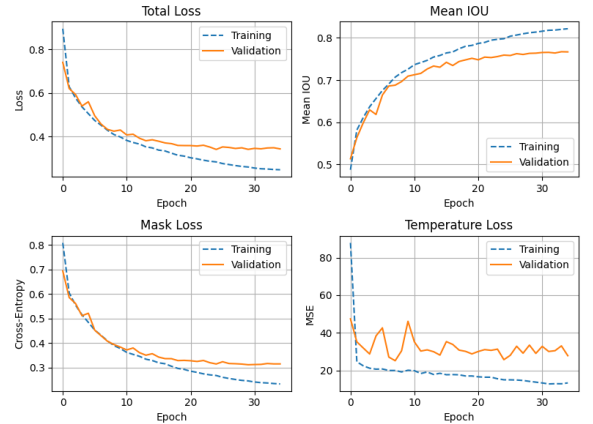


Figure 5: Loss curves with 0.0001 alpha and gradient clipping. Mean IoU well above 0.7.

The results reflect an interplay between the two primary tasks of semantic segmentation and temperature estimation. The semantic segmentation component, indicated by the mask loss and Mean IOU, demonstrates a consistent learning trajectory with Mean IOU reaching upwards of 0.75 to suggest a good alignment between predicted and ground truth masks. This conveys effective learning and generalization capabilities of the model in differentiating between the classes represented in the images. Notably, the parallel decline of both training and

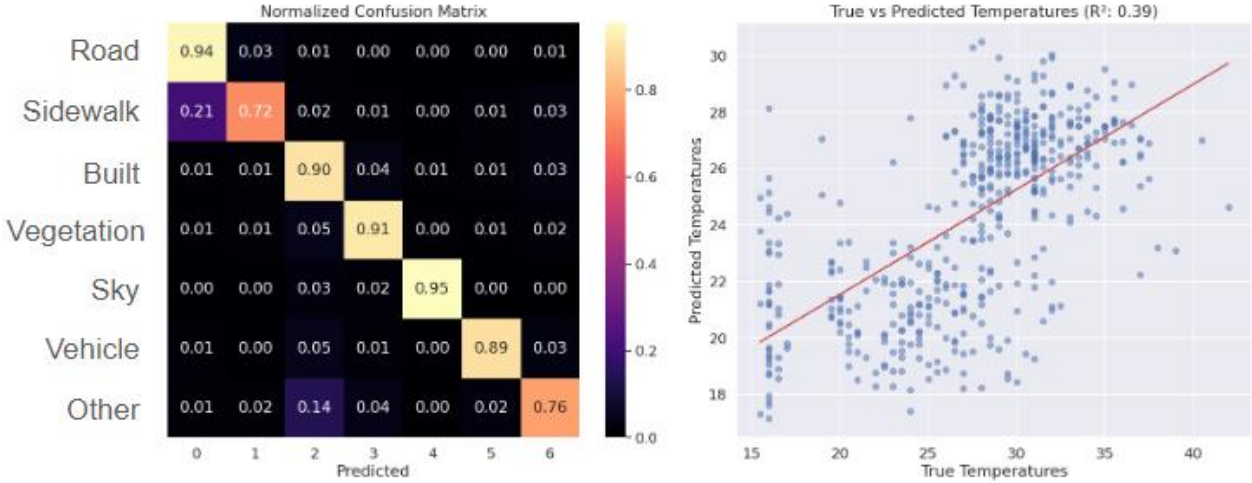


Figure 6: Confusion matrix of the segmentation result and scatterplot of predicted vs true temperature.

validation mask loss suggests an absence of overfitting, corroborating the robustness of the model in segmenting the visual data.

The temperature prediction task, however, exhibits a more complex learning behavior. Initially, there is a marked reduction in the Mean Squared Error (MSE) loss, indicating a positive response to the initial learning rate. A plateau in loss reduction becomes more apparent as more epochs progress. Validation temperature loss plateaus at around MSE of 30, suggesting an RMSE of 5.6 celsius. While the scenario is unwarranted, such a degree of error is acceptable considering the small size of the training dataset.

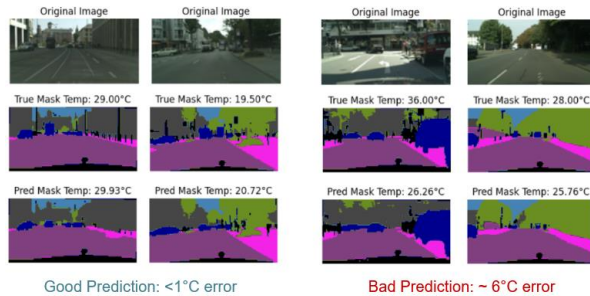


Figure 7: Segmentation result with temperature prediction showing successful and less successful output.

The strong diagonal values in the confusion matrix from Figure 6 represents a relatively high degree of accuracy in the semantic segmentation task. A notable confusion is seen between ‘Sidewalk’ and ‘Road’, as they are similar in structure but differ functionally. The class ‘Other’ also shows a lower accuracy, but that’s expected as it is a conglomerate of different classes with unique features. In

the temperature prediction task, the scatter plot reveals a coefficient of determination (R^2) of 0.39, which indicates a moderate positive correlation between predicted and true temperatures.

Such performance suggests that the model has learned to capture the trend in temperature variation to a certain extent, while variability remains unexplained, highlighting potential avenues for refining the model to better capture the underlying patterns in the temperature data.

4. Conclusion

In this study, we emphasize the dual capabilities of our U-Net based network in both predicting air temperature and segmenting street scene imagery. The training on a diverse dataset of 2975 images yielded promising results, particularly in segmentation tasks, evidenced by a validation IoU consistently above 0.7. However, the temperature prediction, with an RMSE of 5.6°C, indicates potential areas for refinement. The challenge lies in balancing hyperparameters to optimize outputs for both segmentation and temperature prediction tasks, marking a crucial focus for future enhancements in multi-task learning networks covering a similar range of applications.

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